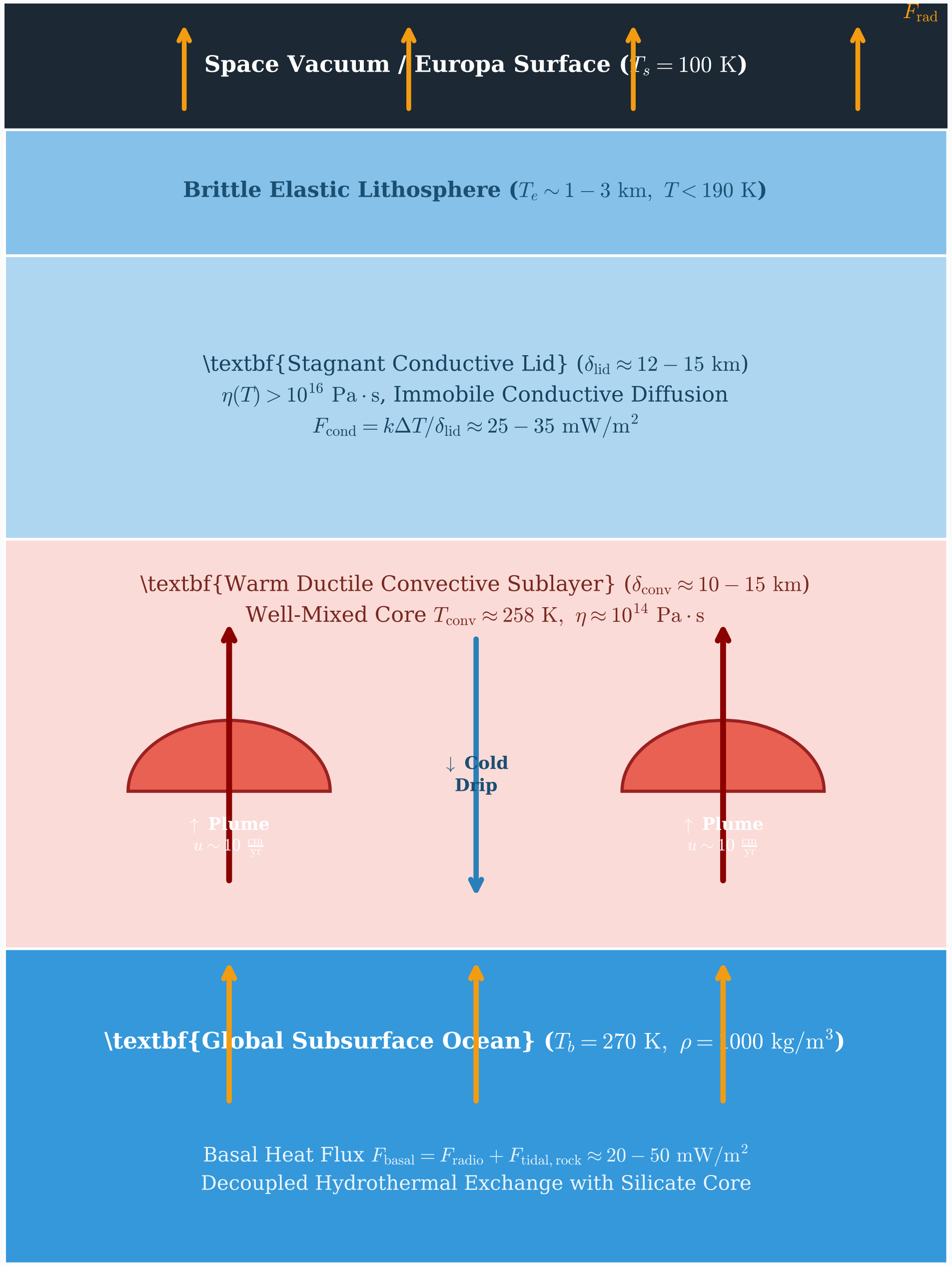
\textbf{Europa Ice Shell Thermal Convection \& Bistability Architecture}

Mitri \& Showman (2005) \textit{Icarus} 177, 447--460 | First-Principles Geophysical Cross-Section



\textbf{I. Convective--Conductive Regime Criteria}

- \textbf{Basal Rayleigh Number}: $Ra_b = \frac{\rho g \alpha \Delta T D^3}{\kappa \eta_{\mathrm{base}}}$
- \textbf{Frank-Kamenetskii Rheology}: $\theta = \frac{E^* \Delta T}{R_g T_b^2} \approx 14.02$
- \textbf{Critical Rayleigh Threshold}: $Ra_{\mathrm{cr}} \approx 20.0\,\theta^4 \approx 7.74 \times 10^5$
- \textbf{Critical Shell Thickness}: $D_{\mathrm{cr}} = \left(\frac{Ra_{\mathrm{cr}} \kappa \eta_{\mathrm{base}}}{\rho g \alpha \Delta T}\right)^{1/3} \approx 14.3$ km
- \textbf{Nusselt Scaling}: $Nu = 0.95\,\theta^{-1.22}\,Ra_b^{0.22} \approx 2-5$ in stagnant lid

\textbf{II. Bistability \& Dynamic Switching Cycle}

\textbf{Conductive-Convective Hysteresis Window} ($F_b \in [15, 42]$ mW/m^2):

1. \textbf{Thick Convective Branch} ($D \approx 20-35$ km):
   High heat transport ($Nu > 2.5$) removes basal flux + tidal heat.
   Sustained by vigorous ductile sublayer convection.

2. \textbf{Thin Conductive Branch} ($D \approx 8-14$ km $< D_{\mathrm{cr}}$):
   Subcritical Rayleigh number suppresses convection.
   Steep thermal gradient conducts heat without flow.

3. \textbf{Perturbation Transitions}:
   - Heat surge triggers rapid catastrophic melting (10^5 yr).
   - Heat deficit causes slow conductive freeze-out (10^6 yr).